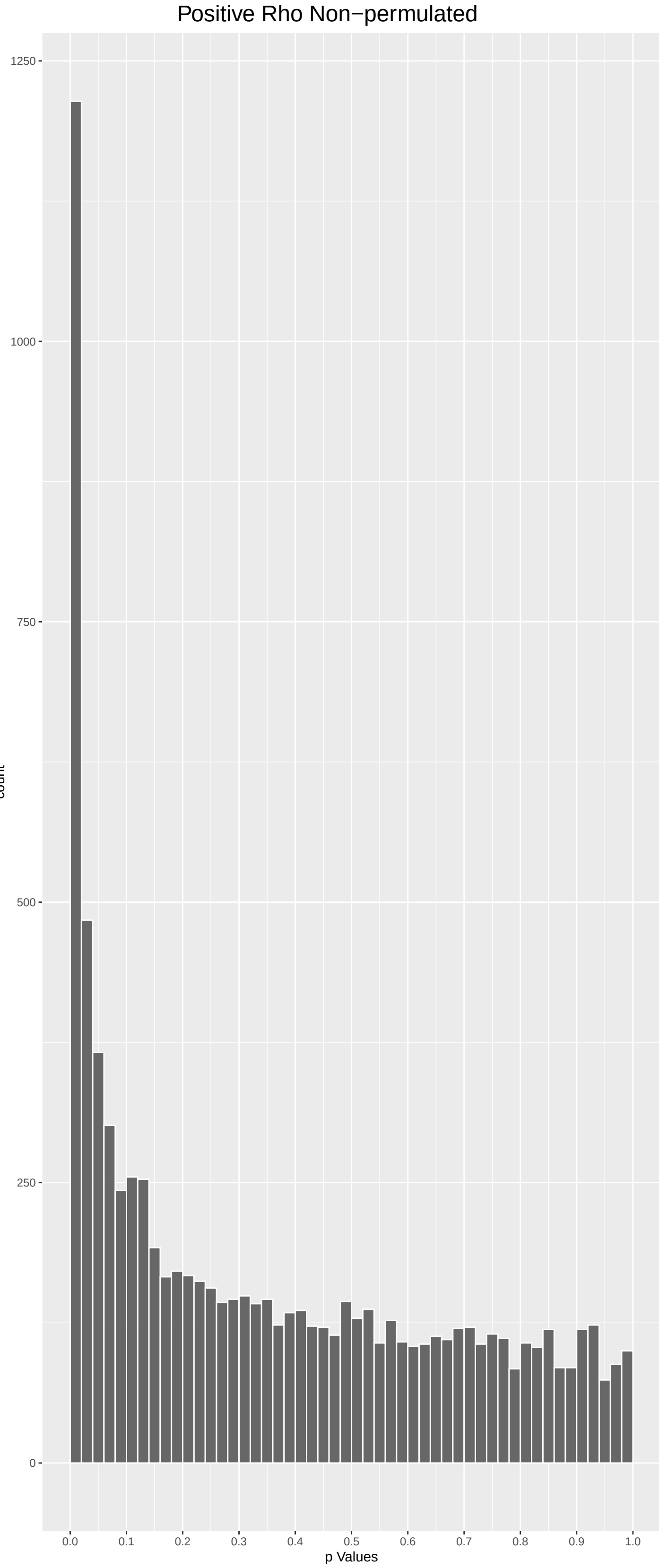


Top genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
LRIT1	-6.215834	5.105283e-10	7.715e-06	1.166e-01
PDE6B	-5.720891	1.059671e-08	7.449e-05	3.449e-01
TRPM6	5.664021	1.478666e-08	7.449e-05	3.449e-01
CPLX3	-5.579183	2.416515e-08	9.130e-05	3.449e-01
CPB1	5.447783	5.100159e-08	1.541e-04	4.659e-01
GALNT5	5.233928	1.659453e-07	3.297e-04	5.028e-01
JAK3	5.224622	1.745111e-07	3.297e-04	5.028e-01
CFAP20DC	5.269109	1.370875e-07	3.297e-04	5.028e-01
TLE6	5.140345	2.742349e-07	4.363e-04	5.028e-01
AKNA	5.130692	2.886793e-07	4.363e-04	5.028e-01
CRYAA2	-5.110648	3.210557e-07	4.411e-04	5.028e-01
TRPM1	-5.056589	4.268215e-07	5.375e-04	5.028e-01
AFF1	5.022187	5.108648e-07	5.939e-04	5.028e-01
SMC4	4.997917	5.795298e-07	6.256e-04	5.028e-01
TASOR2	4.974423	6.544210e-07	6.391e-04	5.028e-01
ABCC10	4.967956	6.766243e-07	6.391e-04	5.028e-01
SPG11	4.949405	7.444051e-07	6.617e-04	5.028e-01
DHDH	4.904048	9.388176e-07	6.987e-04	5.028e-01
PHYH	4.897454	9.708621e-07	6.987e-04	5.028e-01
A2M	4.907683	9.215842e-07	6.987e-04	5.028e-01
THADA	4.921144	8.603977e-07	6.987e-04	5.028e-01
ACADM	4.851257	1.226815e-06	8.239e-04	5.208e-01
ADGRF1	4.846910	1.253990e-06	8.239e-04	5.208e-01
ADAMTS3	4.824613	1.402756e-06	8.833e-04	5.208e-01
MUC20	4.762165	1.915271e-06	1.082e-03	5.208e-01

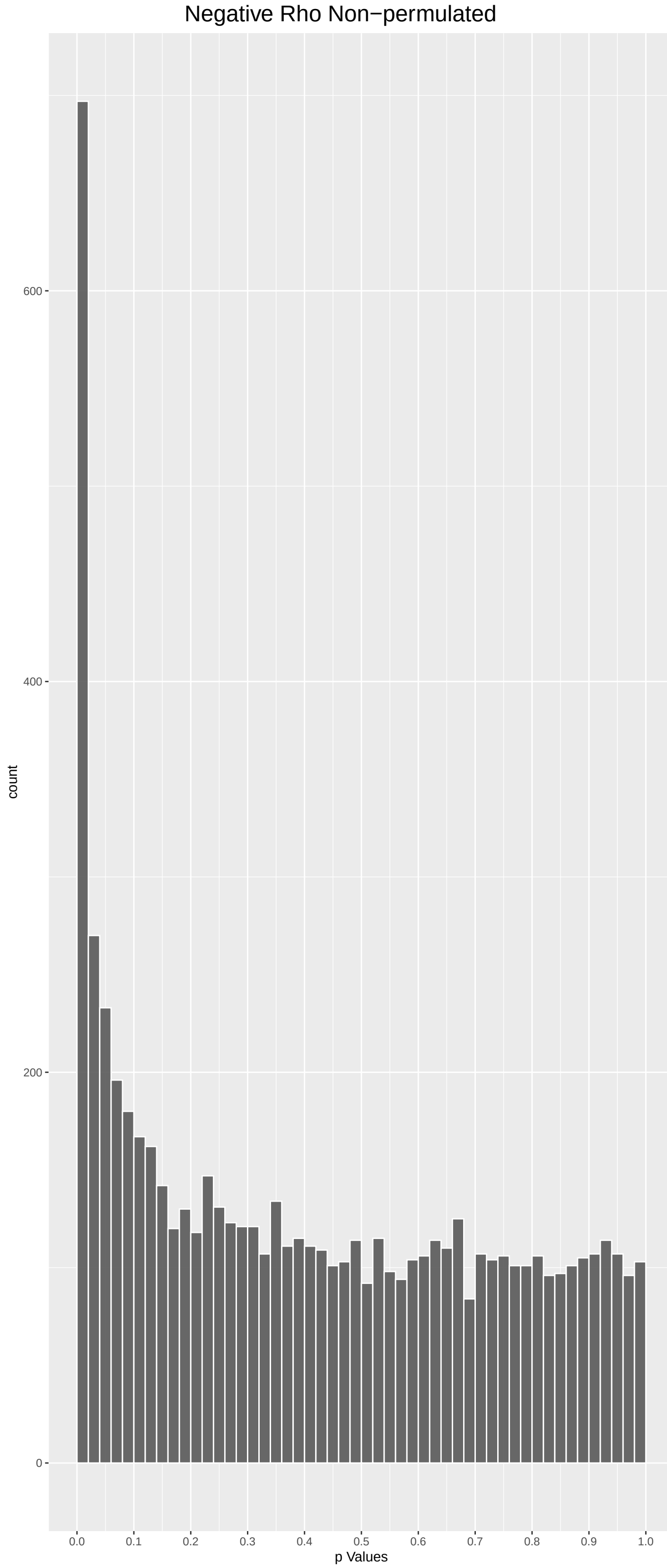


Top Positive genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
TRPM6	5.664021	1.478666e-08	7.449e-05	3.449e-01
CPB1	5.447783	5.100159e-08	1.541e-04	4.659e-01
GALNT5	5.233928	1.659453e-07	3.297e-04	5.028e-01
JAK3	5.224622	1.745111e-07	3.297e-04	5.028e-01
CFAP20DC	5.269109	1.370875e-07	3.297e-04	5.028e-01
TLE6	5.140345	2.742349e-07	4.363e-04	5.028e-01
AKNA	5.130692	2.886793e-07	4.363e-04	5.028e-01
AFF1	5.022187	5.108648e-07	5.939e-04	5.028e-01
SMC4	4.997917	5.795298e-07	6.256e-04	5.028e-01
TASOR2	4.974423	6.544210e-07	6.391e-04	5.028e-01

Top genes by Q-Value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
LRIT1	-6.215834	5.105283e-10	7.715e-06	1.166e-01
PDE6B	-5.720891	1.059671e-08	7.449e-05	3.449e-01
TRPM6	5.664021	1.478666e-08	7.449e-05	3.449e-01
CPLX3	-5.579183	2.416515e-08	9.130e-05	3.449e-01
CPB1	5.447783	5.100159e-08	1.541e-04	4.659e-01
DHDH	4.904048	9.388176e-07	6.987e-04	5.028e-01
TLE6	5.140345	2.742349e-07	4.363e-04	5.028e-01
GALNT5	5.233928	1.659453e-07	3.297e-04	5.028e-01
SPG11	4.949405	7.444051e-07	6.617e-04	5.028e-01
PHYH	4.897454	9.708621e-07	6.987e-04	5.028e-01
A2M	4.907683	9.215842e-07	6.987e-04	5.028e-01
TASOR2	4.974423	6.544210e-07	6.391e-04	5.028e-01
SMC4	4.997917	5.795298e-07	6.256e-04	5.028e-01
ABCC10	4.967956	6.766243e-07	6.391e-04	5.028e-01
AKNA	5.130692	2.886793e-07	4.363e-04	5.028e-01
AFF1	5.022187	5.108648e-07	5.939e-04	5.028e-01
THADA	4.921144	8.603977e-07	6.987e-04	5.028e-01
JAK3	5.224622	1.745111e-07	3.297e-04	5.028e-01
CFAP20DC	5.269109	1.370875e-07	3.297e-04	5.028e-01
TRPM1	-5.056589	4.268215e-07	5.375e-04	5.028e-01
CRYAA2	-5.110648	3.210557e-07	4.411e-04	5.028e-01
ADAMTS3	4.824613	1.402756e-06	8.833e-04	5.208e-01
SUSD5	4.713827	2.431074e-06	1.155e-03	5.208e-01
MUC20	4.762165	1.915271e-06	1.082e-03	5.208e-01
TDRD7	-4.685420	2.793860e-06	1.206e-03	5.208e-01



Top Negative genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
LRIT1	-6.215834	5.105283e-10	7.715e-06	1.166e-01
PDE6B	-5.720891	1.059671e-08	7.449e-05	3.449e-01
CPLX3	-5.579183	2.416515e-08	9.130e-05	3.449e-01
CRYAA2	-5.110648	3.210557e-07	4.411e-04	5.028e-01
TRPM1	-5.056589	4.268215e-07	5.375e-04	5.028e-01
CRYBB2	-4.755112	1.983368e-06	1.082e-03	5.208e-01
CHST15	-4.718289	2.378366e-06	1.155e-03	5.208e-01
OR10D3	-4.692113	2.703982e-06	1.202e-03	5.208e-01
TDRD7	-4.685420	2.793860e-06	1.206e-03	5.208e-01
KHDRBS3	-4.636818	3.538134e-06	1.445e-03	5.902e-01

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Modulation Of Chemical Synaptic Transmis	-0.20569709	105	3.636e-13	1.959e-09	CPLX3:3 DRD2:28 DGKI:44 TRIO:52 GRM6:85 NPTX1:88
NADH Dehydrogenase Complex Assembly (GO: 0.29380758	0.29380758	48	1.969e-12	3.536e-09	NDUF88:76 ECSIT:170 NDUF9A:278 TMEM1268:302 NDUF86:335 TMEM126A:366
Mitochondrial Respiratory Chain Complex	0.29380758	48	1.969e-12	3.536e-09	NDUF88:76 ECSIT:170 NDUF9A:278 TMEM1268:302 NDUF86:335 TMEM126A:366
Chemical Synaptic Transmission (GO:00072	-0.12974537	237	7.453e-12	1.004e-08	GABRR1:14 DRD2:28 HRH3:53 RPS6KA2:63 NPTX1:88 SLC18A2:95
Mitochondrial Respiratory Chain Complex	0.21577959	77	1.642e-11	6.749e-08	NDUF88:76 ECSIT:170 NDUF9A:278 TMEM1268:302 NDUF86:335 TMEM126A:366
Mitochondrial Gene Expression (GO:014005	0.18623238	99	1.642e-10	1.474e-07	PTCD3:56 TFB2M:98 POLRM1:128 TFAM:342 MRPL37:412 MRPS30:469
Visual Perception (GO:0007801)	-0.18494442	90	1.410e-09	1.085e-06	TRPM1:5 CRYBB2:6 CRYAA:12 CRYBB1:40 VAX2:45 CRYBB3:66
Calcium Ion Transport (GO:0006816)	-0.17007601	102	3.144e-09	2.117e-06	TRPM1:5 TRPM5:35 ATP2A2:67 CACNA2D4:112 ITPR1:150 TRPM7:229
Mitochondrial Electron Transport, NADH T	0.30232565	31	5.777e-09	3.458e-06	NDUF88:76 NDUF9A:278 NDUFV3:281 NDUF86:335 NDUF6A:471 NDUF85:473
Mitochondrial ATP Synthesis coupled Elec	0.23016349	52	9.666e-09	3.759e-06	NDUF88:76 NDUF9A:278 NDUFV3:281 NDUF86:335 NDUF6A:471 NDUF85:473
Monoatomic Cation Transmembrane Transport	-0.10240286	269	8.810e-09	3.759e-06	TRPM1:5 TRPM5:35 KCNA10:51 ATP2A2:67 KCNC2:78 KCNQ3:100
Oxidative Phosphorylation (GO:0006119)	0.22604293	54	9.433e-09	3.759e-06	NDUF88:76 ATP5PO:105 NDUF9A:278 NDUFV3:281 NDUF86:335 NDUF6A:471
Proton Motive Force-Driven ATP Synthesis	0.23232710	51	9.768e-09	3.759e-06	NDUF88:76 ATP5PO:105 NDUF9A:278 NDUFV3:281 NDUF86:335 NDUF6A:471
Sensory Perception Of Light Stimulus (GO	-0.17530134	91	7.920e-09	3.759e-06	TRPM1:5 CRYBB2:6 CRYAA:12 CRYBB1:40 VAX2:45 CRYBB3:66
Anterograde Trans-Synaptic Signaling (GO	-0.12720026	169	1.296e-08	4.493e-06	GABRR1:14 HRH3:53 RPS6KA2:63 NPTX1:88 SLC18A2:95 KCNQ3:100
Mitochondrial Translation (GO:0032543)	0.16896726	95	1.334e-08	4.493e-06	PTCD3:56 MRPL37:412 MRPS30:469 GATB:476 MRPS23:559 MRPL9:732
Proton Motive Force-Driven Mitochondrial	0.23709897	47	1.918e-08	6.077e-06	NDUF88:76 ATP5PO:105 NDUF9A:278 NDUFV3:281 NDUF86:335 NDUF6A:471
Muscle Contraction (GO:0006936)	-0.17530663	84	2.885e-08	8.634e-06	MYH13:198 NOS1:341 MYOM2:344 LMFD2:407 DES:443 KCN12:469
Aerobic Electron Transport Chain (GO:001	0.22086990	52	3.695e-08	1.039e-05	NDUF88:76 NDUF9A:278 NDUFV3:281 NDUF86:335 NDUF6A:471 NDUF85:473
Potassium Ion Transmembrane Transport (G	-0.13996686	130	3.856e-08	1.039e-05	KCNA10:51 KCNC2:78 KCNQ3:100 KCNQ2:125 KCNS3:129 KCNJ5:168
Regulation Of Synaptic Transmission, Glu	-0.21493221	47	3.508e-07	8.998e-05	DRD2:28 DGKI:44 GRM6:85 MAPKB8:132 TSHZ3:179 GRM8:204
tRNA Methylation (GO:0030488)	0.23999558	36	6.347e-07	1.554e-04	THADA:13 TRMT1:91 TRMT10A:210 FTSJ1:451 TRMT10C:645 TRMT9B:674
Alpha-Amino Acid Catabolic Process (GO:1	0.26535464	29	7.662e-07	1.794e-04	HIBCH:324 HMGCL:417 ALDH4A1:456 TAT:1107 KMO:1179 GOT2:1238
Aerobic Respiration (GO:0009060)	0.19580357	53	8.367e-07	1.878e-04	NDUF88:76 NDUF9A:278 NDUFV3:281 NDUF86:335 NDUF6A:471 NDUF85:473
Positive Regulation Of Synaptic Transmis	-0.17627688	65	9.153e-07	1.972e-04	SHANK2:11 TSHZ3:179 CACNG5:205 PTKB2:373 SYT1:398 STX4:87
Potassium Ion Transport (GO:0006813)	-0.13042014	116	1.288e-06	2.668e-04	KCNA10:51 KCNC2:78 KCNQ3:100 KCNQ2:125 KCNS3:129 KCNJ5:168
Inorganic Cation Transmembrane Transport	-0.08518007	272	1.513e-06	3.019e-04	TRPM1:5 TRPM5:35 KCNA10:51 ATP2A2:67 KCNC2:78 KCNQ3:100
Calcium Ion Transmembrane Transport (GO:	-0.15677935	78	1.750e-06	3.367e-04	TRPM1:5 TRPM5:35 ATP2A2:67 CACNA2D4:112 TRPM7:229 LETM1:411
Skeletal Muscle Contraction (GO:0006941)	-0.19380745	50	2.169e-06	4.030e-04	TNNI2:222 CSRP3:322 NOS1:341 RYR2:505 TNNI1:922 TCAP:939
KEGG Conduction (GO:0006137)	-0.21243957	41	2.555e-06	4.588e-04	BIN1:124 KCNS3:168 DSC2:190 CAMK2D:287 DSG2:430 MEF2A:687
Metal Ion Transport (GO:0030001)	-0.11194512	148	2.764e-06	4.803e-04	KCNA10:51 KCNS3:129 ITPR1:150 KCNJ5:168 TRPM7:229 SCN2A:240
Fatty Acid Oxidation (GO:0019395)	0.19710159	47	2.993e-06	4.885e-04	PHYH:16 ACADM:17 CPT2:116 HAO2:207 EHHADH:216 EC12:304
Mitochondrial RNA Metabolic Process (GO:	0.30967997	19	2.982e-06	4.885e-04	TFB2M:98 POLRM1:128 TFAM:342 TEFM:507 FASTKD5:648 FASTKD1:713
Fatty Acid Beta-Oxidation (GO:0006635)	0.19652999	47	3.196e-06	5.065e-04	ACADM:17 ACOX2:38 CPT2:116 EHHADH:216 EC12:304 GCDH:378
tRNA Modification (GO:0006400)	0.16444828	66	3.936e-06	6.057e-04	THADA:13 TRMT1:91 TRMT10A:210 FTSJ1:451 DTWD1:532 TRMT9B:674
Calcium Ion Import Across Plasma Membran	-0.22732193	33	6.269e-06	9.293e-04	TRPM1:5 CACNA1B:197 SCN2A:240 CACNA1F:451 SLC24A4:460 SLC24A2:882
Double-Strand Break Repair (GO:0006302)	0.10850866	146	6.383e-06	9.293e-04	ERCC5:32 RNF169:53 ANKLE1:65 RNF168:136 REC8:269 BRCA1:275
Cell Junction Assembly (GO:0034329)	-0.14521986	80	7.328e-06	1.014e-03	SHANK2:11 DRD2:28 SKD1:96 ITGA6:188 DBNL:249 SLITRK1:272
Fatty Acid Catabolic Process (GO:0009062	0.17038062	58	7.338e-06	1.014e-03	PHYH:16 ACADM:17 CPT2:116 HAO2:207 EHHADH:216 EC12:304
Monoatomic Ion Transport (GO:0006811)	-0.12768593	103	7.855e-06	1.058e-03	SLC15A1:194 SLC12A5:256 CNGB3:274 CHRNA4:285 SLC24A4:460 TRPM3:465

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_NEURONAL_SYSTEM	-0.15397617	352	4.750e-23	3.081e-19	SHANK2:11 GABRR1:14 ADCY1:39 NBEA:48 KCNA10:51 RPS6KA2:63
REACTOME_TRANSMISSION_ACROSS_CHEMICAL_SY	-0.14559047	217	1.652e-13	5.359e-10	GABRR1:14 ADCY1:39 NBEA:48 RPS6KA2:63 PRKCG:91 SLC18A2:95
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	0.30619674	47	3.873e-13	8.375e-10	NDUF88:76 ECSIT:170 NDUFV3:281 TMEM1268:302 NDUF86:335 NDUF6A:471
REACTOME_NEUROTRANSMITTER_RECEPTORS_AND	-0.16609748	159	5.362e-13	8.696e-10	GABRR1:14 ADCY1:39 NBEA:48 RPS6KA2:63 PRKCG:91 CAMK2B:110
REACTOME_COMPLEX_I_BIOGENESIS	0.29877596	48	8.144e-13	1.057e-09	NDUF88:76 ECSIT:170 NDUF9A:278 NDUFV3:281 TMEM1268:302 NDUF86:335
REACTOME_MUSCLE_CONTRACTION	-0.15267498	176	3.076e-12	3.326e-09	ATP2A2:67 CAMK2B:110 ITPR1:150 TNNI2:222 SCN2A:240 CAMK2D:287
REACTOME_DEVELOPMENTAL_BIOLOGY	-0.06865569	792	7.478e-11	6.930e-08	DSG3:27 COL3A1:50 TRIO:52 PTK2:54 L1CAM:57 DOCK1:61
KEGG_CALCIIUM_SIGNALING_PATHWAY	-0.14673418	161	1.416e-10	1.149e-07	ADCY1:39 ATP2A2:67 PRKCG:91 P2RX2:107 CAMK2B:110 ITPR1:150
REACTOME_INFECTIOUS_DISEASE	-0.07231500	671	2.305e-10	1.523e-07	ADCY1:39 PTK2:54 DOCK1:61 DYNCH1H:80 ACE2:84 ITPR1:150
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	0.20636552	79	2.348e-10	1.523e-07	NDUF88:76 LRPPRC:106 ECSIT:170 NDUF9A:278 NDUFV3:281 TMEM1268:302
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT_	0.17825273	99	9.810e-10	5.414e-07	NDUF88:76 ATP5PO:105 LRPPRC:106 ECSIT:170 NDUF9A:278 NDUFV3:281
REACTOME_GENE_SILENCING_BY_RNA	-0.19234504	82	1.781e-09	9.627e-07	MOV10L1:20 PIWIL1:36 FKBP6:59 TNRC6B:147 ASZ1:237 DICER1:339
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	-0.07730675	514	2.454e-09	1.225e-06	DRD2:28 CADM3:55 NETO1:58 CPZ:60 FAM163B:65 GPR26:79
NIKOLSKY_BREAST_CANCER_7P22_AMPLICON	-0.27817786	38	2.985e-09	1.300e-06	TTYH3:138 MAFK:184 INTS1:236 LFNMG:332 GET4:413 EIF3B:518
KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERAC	-0.10713848	260	3.006e-09	1.300e-06	GABRR1:14 DRD2:28 MAS1:33 HRH3:53 GRM6:85 P2RX2:107
REACTOME_NERVOUS_SYSTEM_DEVELOPMENT	-0.08141721	436	6.583e-09	2.669e-06	COL3A1:50 TRIO:52 PTK2:54 L1CAM:57 DOCK1:61 RPS6KA2:63
WP_CALCIIUM_REGULATION_IN_CARDIAC_CELLS	-0.15652764	114	8.079e-09	3.083e-06	ADCY1:39 ATP2A2:67 GJA8:75 PRKCG:91 CAMK2B:110 ITPR1:150
NIKOLSKY_BREAST_CANCER_8P12_P11_AMPLICON	0.23485170	50	9.317e-09	3.358e-06	NSD3:59 IDO2:60 RAB11FIP1:73 ADAM9:103 DKK4:153 TACCL1:161
REACTOME_PROCESSING_OF_CAPPED_INTRON_CON	-0.12067316	186	1.472e-08	5.025e-06	SLBP:116 RBM42:161 CPSF7:207 RPKM2:275 NCBP1:292 PAPOLA:293
BENPORATH_ES_WITH_H3K27ME3	-0.05541736	913	1.932e-08	5.967e-06	NKPD1:24 HSFA4:42 DGKI:44 VAX2:45 ENTPD2:49 IGFBP1:62
WP_OXIDATIVE_PHOSPHORYLATION	0.23955179	46	1.918e-08	5.967e-06	NDUF88:76 ATP5PO:105 NDUF9A:278 NDUFV3:281 NDUF86:335 NDUF6A:471
REACTOME_KERATINIZATION	-0.17802238	83	2.216e-08	6.238e-06	DSG3:27 DSC2:190 KRT20:218 DSC3:259 KLK13:277 PKP3:423
REACTOME_G_ALPHA_I_SIGNALING_EVENTS	-0.10404111	245	2.218e-08	6.256e-06	ADCY1:39 NBEA:48 GRM6:85 PRKCG:91 GNAT3:293 OPN3:108
BENPORATH_EED_TARGETS	-0.05648250	860	2.578e-08	6.967e-06	NKPD1:24 HSFA4:42 DGKI:44 VAX2:45 KCNC2:78 SORCS2:86
REACTOME_GABA_RECEPTOR_ACTIVATION	-0.24585341	42	3.561e-08	9.240e-06	GABRR1:14 ADCY1:39 GABRB3:146 KCNJ5:168 KCNJ9:169 ADCY9:348
REACTOME_SIGNALING_BY_GPCR	-0.06629236	596	3.981e-08	9.933e-06	TIAM2:26 DRD2:28 ADCY1:39 DGKI:44 NBEA:48 TRIO:52
WP_2Q37_COPY_NUMBER_VARIATION_SYNDROME	-0.15526161	104	4.619e-08	1.110e-05	KIF1A:72 ASB1:128 KLHL30:181 TRAF3IP1:187 GPC1:317 HADC4:438
REACTOME_CARDIAC_CONDUCTION	-0.14789741	113	5.793e-08	1.342e-05	ATP2A2:67 CAMK2B:110 ITPR1:150 SCN2A:240 CAMK2D:287 NOS1:341
WONG_MITOCHONDRIA_GENE_MODULE	0.11353669	192	6.163e-08	1.379e-05	CHDH:49 NDUF88:76 ATP5PO:105 LRPPRC:106 CRYZ:107 SLC16A5:237
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	-0.14275234	120	6.860e-08	1.483e-05	HRH3:53 ZBTB46:106 SAMD10:9 KCNQ12:125 OPRL1:175 TCEA2:247
BLANCO_MELO_BRONCHIAL_EPITHELIAL_CELLS_I	0.12363118	159	7.833e-08	1.612e-05	KIF18B:85 FANGC:160 CLSPN:163 SPAG5:176 ZWINT:187 UHRF1:190
REACTOME_ION_HOMEOSTASIS	-0.23137330	45	7.951e-08	1.612e-05	ATP2A2:67 CAMK2B:110 ITPR1:150 CAMK2D:287 NOS1:341 ATP2B3:492
BENPORATH_SUZ12_TARGETS	-0.05458051	852	8.466e-08	1.646e-05	HSF4:42 DGKI:44 VAX2:45 CADM3:55 KCNC2:78 EGLFAM:81
REACTOME_MITOCHONDRIAL_TRANSLATION	0.15907454	91	1.603e-07	3.001e-05	PTCD3:56 MRPL37:412 MRPS30:469 MRPS23:559 MRPL9:732 MTIF2:831
REACTOME_FORMATION_OF_THE_CORNIFIED_ENVE	-0.17391663	76	1.619e-07	3.001e-05	DSG3:27 DSC2:190 KRT20:218 DSC3:259 KLK13:277 PKP3:423
MEISSNER_NPC_HCP_WITH_H3KAME2_AND_H3K27M	-0.06673217	308	1.815e-07	3.271e-05	HRH3:53 CADM3:55 FAM163B:65 EGLFAM:81 ADGBR1:89 KCNQ2:125
REACTOME_HCMV_EARLY_EVENTS	-0.16222928	86	2.034e-07	3.566e-05	DYNCH1H:80 NUP37:400 TUBA1B:555 NUP43:620 NUP160:699 NUP133:768
KIM_ALL_DISORDERS_CALB1_CORR_UP	-0.07230861	442	2.143e-07	3.659e-05	KHDRBS3:10 ATP2A2:67 DYNCH1H:80 NPTX1:88 CAMK2B:110 DCLK1:142
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON	-0.37959733	15	3.579e-07	5.805e-05	EP400:23 P2RX2:107 SFSWAP:223 POLE:380 GOLGA3:435 GALNT9:551
REACTOME_VOLTAGE_GATED_POTASSIUM_CHANNEL	-0.22983191	41	3.568e-07	5.805e-05	KCNA10:51 KCNC2:78 KCNQ3:100 KCNQ2:125 KCNS3:129 KCNV2:234

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Schizophrenia	-0.05243317	1513	3.933e-11	3.857e-07	TRPM1:5 SHANK2:11 GABRR1:14 DRD2:28 ACTL7A:31 ADCY1:39
Mitochondrial Diseases	0.10374713	335	8.837e-11	4.332e-07	NDUF88:76 LRPPRC:106 CRYZ:107 CPT2:116 ECSIT:170 CEP95:222
Autism Spectrum Disorders	-0.07595925	456	1.100e-08	3.595e-05	SHANK2:11 DRD2:28 JAKMIP1:46 NBEA:48 TRIO:52 L1CAM:57
Increased CSF lactate	0.20159533	53	3.923e-07	7.825e-04	LRPPRC:106 NDUF9A:278 COX10:286 TMEM1268:302 NDUF53:597 NDUF56:625
Nuclear cataract	-0.25761561	32	4.624e-07	7.825e-04	CRYBB2:6 CRYAA:12 CRYBB1:40 HSFA4:42 CRYBB3:66 GJA8:75
Primary microcephaly	0.14817066	97	4.788e-07	7.825e-04	ERCC5:32 FANGC:160 TRMT10A:210 FANCE:239 CENP1:265
CATARACT, AUTOSOMAL DOMINANT	-0.44923205	10	8.706e-07	1.219e-03	CRYAA2:4 CRYBB2:6 CRYAA:12 CRYBB1:40 GJA8:75 MIP:126
Cataract, Nucleus	-0.46800229	9	1.164e-06	1.427e-03	CRYBB2:6 CRYBB1:40 HSFA4:42 GJA8:75 BFPSP2:215 VIM:395
Difficulties with night vision	-0.13298068	109	1.689e-06	1.840e-03	PDE6B:2 TRPM1:5 CDHR1:15 NR2E3:74 GRM6:85 CACNA2D4:112
Abnormality of macular pigmentation	-0.29578100	21	2.718e-06	2.220e-03	PDE6B:2 TRPM1:5 GRM6:85 CACNA2D4:112 PRPH2:193 ABCA4:208
Macular pigmentary changes	-0.29578100	21	2.718e-06	2.220e-03	PDE6B:2 TRPM1:5 GRM6:85 CACNA2D4:112 PRPH2:193 ABCA4:208
Nuclear non-senile cataract	-0.25278563	29	2.480e-06	2.220e-03	CRYBB2:6 CRYAA:12 CRYBB1:40 CRYBB3:66 GJA8:75 MIP:126
Comatose	0.15935214	71	3.528e-06	2.471e-03	ACADM:17 CPT2:116 TMEM1268:302 HMGCL:417 ABCD2:517 NDUF53:597
Night Blindness	-0.12608336	114	3.466e-06	2.471e-03	PDE6B:2 TRPM1:5 CDHR1:15 NR2E3:74 GRM6:85 CACNA2D4:112
Congenital cataract	-0.14135643	88	4.726e-06	3.089e-03	CRYAA2:4 CRYBB2:6 CRYAA:12 CRYBB1:40 HSFA4:42 CRYBB3:66
CATARACT, COPOCK-LIKE	-0.43910381	9	5.081e-06	3.114e-03	CRYBB2:6 CRYBB1:40 GJA8:75 BFPSP2:215 VIM:395 GJA3:1031
Channeopathies	-0.19573689	45	5.630e-06	3.247e-03	KCNJ5:168 CNGB3:274 KCNJ6:374 CACNA1F:451 RYR2:505 SHANK3:572
Mouth Neoplasms	-0.12981944	99	8.353e-06	4.550e-03	PTK2:54 PTH1R:178 VIM:395 TIAM1:459 MGMT:511 HMGC2:515
Severe myopia	-0.13528396	88	1.187e-05	6.126e-03	PDE6B:2 TRPM1:5 GRM6:85 CACNA2D4:112 CABP4:235 CNGB3:274
Night blindness, congenital stationary	-0.26218851	23	1.352e-05	6.446e-03	PDE6B:2 TRPM1:5 USP11:30 GRM6:85 CACNA2D4:112 ABCA4:208
Alzheimer's Disease	-0.03413572	1551	1.381e-05	6.446e-03	SHANK2:11 NKPD1:24 DRD2:28 PLD1:34 COL3A1:50 HRH3:53
NADH:Q(1) Oxidoreductase deficiency	0.24879551	25	1.547e-05	6.894e-03	TMEM1268:302 NDUF53:597 NDUF56:625 FOXRED1:638 NDUF9A:2701 TIMMDC1:858
Congenital myopathy (disorder)	-0.12369091	99	2.180e-05	8.907e-03	COL12A1:92 BIN1:124 TNNI2:222 COL6A2:372 HADC1